Week 96: Prime bags

Adult guide built from the problem files · Unpiloted; kits and timing unrehearsed.

The answer keys, proofs, preparation and launch are in the print guide, week-96-facilitator.pdf, whose source is guide/facilitator.tex. This view lists each problem's numbers, route and idea.

Problems and routes

Problem	2–3	4–5	route
Different splitting routes, one prime bag?	P1	P1	first
Every value a bag can make	P2	P2	first
Four tokens against three	P3		first
Count the values without listing them		P3	first
Seven 2s and four 3s		P4	first
Bags that make exactly 4, 6 or 8 values	P4		return
Token-count patterns for a divisor count		P5	return
The least number with each divisor count		P6	return

First route. Grades 2–3 P1–P3; Grades 4–5 P1–P4. **Return visits.** Grades 2–3 P4; Grades 4–5 P5–P6.

Key, one entry per problem

Different splitting routes, one prime bag?

Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Every splitting route of a number ends in the same bag of primes: unique factorization, met as a conjecture.

Every value a bag can make

Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. The values made by choosing tokens are exactly the divisors of the full bag's value; the empty bag gives 1.

Four tokens against three

Grades 2–3 P3 · first route

Idea. $24 = 2\cdot2\cdot2\cdot3$ and $30 = 2\cdot3\cdot5$ both have 8 divisors: the number of tokens does not decide the count.

Count the values without listing them

Grades 4–5 P3 · first route

Idea. Choosing how many of each prime gives (count + 1) choices per prime, so 36, 60 and 72 have 9, 12 and 12 divisors.

Seven 2s and four 3s

Grades 4–5 P4 · first route

Idea. A bag with counts $e_1, \dots, e_r$ has $(e_1 + 1) \cdots (e_r + 1)$ divisors; seven 2s and four 3s give 40.

Bags that make exactly 4, 6 or 8 values

Grades 2–3 P4 · return route

Idea. The smallest bags with 4, 6 and 8 values are 6, 12 and 24.

Token-count patterns for a divisor count

Grades 4–5 P5 · return route

Idea. The patterns for a target are its factorizations into (count + 1) factors; 12 has four.

The least number with each divisor count

Grades 4–5 P6 · return route

Idea. Put the largest counts on the smallest primes: 6, 12, 24, 36, 48 and 60 for 4, 6, 8, 9, 10 and 12 divisors.